1 If incorrect, award [B1] for any two of 12, x^8, or y^4 correct in a product

$12x^8y^4$ **[B2]**

[Total Marks: 2]

2 **(a)** $12x^{-3}$ or $\dfrac{12}{x^3}$ **[B1]**

(b) $\dfrac{2x^3}{3}$ or $\dfrac{2}{3}x^3$ **[B2]**

Correct cube root of coefficient $\dfrac{2}{3}$ OR correct index x^3 seen **[M1]**

[Total Marks: 3]

3 **(a)** $20x$ **[B1]**

(b) Numerator $4x^8$ or denominator y^6 seen **[M1]**

$\dfrac{4x^8}{y^6}$ **[A1]**

(c) -3 **[B1]**

[Total Marks: 4]

4 **(a)** M1 for any two of 12, x^8 or y^4 seen in a product

$12x^8y^4$ **[A1]**

(b) M1 for $\dfrac{1}{16^{\frac{1}{4}}}$ or 2^{-1} or $\dfrac{1}{\sqrt[4]{16}}$

$\dfrac{1}{2}$ **[A1]**

[Total Marks: 4]

5 **(a)** 15 or x^2 in a product **[M1]**

$15x^2$ **[A1]**

(b) $\sqrt[4]{16}$ or 2^3 or $4096^{\frac{1}{4}}$ **[M1]**

8 **[A1]**

[Total Marks: 4]

6 **(a)** y^3 **[B1]**

(b) $\dfrac{1}{25}$ or 0.04 **[B1]**

(c) $8x^{12}$ **[B2]**

If B0, allow [M1] for $8x^k$ or kx^{12} where $k \neq 0$

[Total Marks: 4]

7 **(a)** Coefficient 6 or power $-3+\frac{1}{2}$ seen **[M1]**

$6x^{-\frac{5}{2}}$ or $\frac{6}{x^{\frac{5}{2}}}$ **[A1]**

(b) Two of 27, y^{12}, or denominator 8 correct **[M1]**

$\frac{27y^{12}}{8}$ **[A1]**

[Total Marks: 4]

8 **(a)** -3 **[B1]**

(b) $2y^3$ **[B2]**

If not B2, then B1 for 2 or y^3 seen **[B1]**

(c) $\frac{1}{8}$ **[B1]**

[Total Marks: 4]

9 **(a)** w^{11} **[B1]**

(b) 3^2 or 9 seen, or x^6 seen **[M1]**

$9x^6$ **[A1]**

(c) $\frac{15}{5}=3$ or $y^{8-(-2)}$ **[M1]**

$3y^{10}$ **[A1]**

[Total Marks: 5]

10 **(a)** p^{11} **[B1]**

(b) $9q^6$ **[B2]**

Partial correct answer $9q^k$ or cq^6 **[B1]**

(c) $(\sqrt[3]{27})^2$ or 3^2 or $\sqrt[3]{729}$ **[M1]**

9 **[A1]**

[Total Marks: 5]

11 **(a)** $4x^3y^4$ **[B2]**

If not B2, then B1 for two of 4, x^3, y^4 correct **[B1]**

(b) $(3^3)^{x-1}=(3^2)^{2x+1}$ or $3^{3(x-1)}=3^{2(2x+1)}$ **[M1]**

$3x-3=4x+2$ **[M1]**

-5 **[A1]**

[Total Marks: 5]

12 **(a)** $\frac{1}{4}$ or 0.25 **[B2]**

Correct intermediate step e.g., 2^{-2} or $\left(\frac{1}{32}\right)^{\frac{2}{5}}$ or $32^{\frac{1}{5}} = 2$ **[M1]**

(b) Writes 125 as 5^3 **[M1]**

Equates indices: $x + 2 = 3x$ **[M1]**

1 **[A1]**

[Total Marks: 5]

13 **(a)** $(\sqrt[3]{125})^2$ or 5^2 or $125^2 = 15625$ and $\sqrt[3]{15625}$ **[M1]**

25 **[A1]**

(b) 3 **[B1]**

x^6 **[B1]**

y^{-8} or $\frac{1}{y^8}$ **[B1]**

Note: Award [B3] for a fully correct final answer of $\frac{3x^6}{y^8}$ or $3x^6y^{-8}$

[Total Marks: 5]

14 **(a)** $\frac{1}{4}$ or 0.25 **[B1]**

(b) 25 or $\frac{1}{25}$ seen **[M1]**

1 **[A1]**

(c) $125 = 5^3$ seen or implied **[B1]**

$2x - 1 = 3x$ **[M1]**

$x = -1$ **[A1]**

[Total Marks: 6]

15 **(a)** $4 \times \frac{3}{2} = 6$ or $x^{-3+\frac{1}{2}}$ **[M1]**

$6x^{-\frac{5}{2}}$ or $\frac{6}{x^{\frac{5}{2}}}$ **[A1]**

(b) $3^{2x+1} = 3^3$ or $2x + 1 = 3$ **[M1]**

1 **[A1]**

(c) $8^{\frac{1}{3}} = 2$ or $(a^6)^{\frac{1}{3}} = a^2$ or $(b^3)^{\frac{1}{3}} = b$ **[M1]**

$\dfrac{2a^2}{b}$ **[A1]**

[Total Marks: 6]

16 (a) Numerator $8x^{15}$ **[B1]**

Denominator $27y^6$ **[B1]**

$\dfrac{8x^{15}}{27y^6}$ **[A1]**

(b) Writing both sides with a common base of 3, e.g. $3^{2x} = 3^{3(x+1)}$ **[M1]**

$2x = 3x + 3$ **[M1]**

-3 **[A1]**

[Total Marks: 6]

17 (a) $\left(\dfrac{64}{27}\right)^{\frac{2}{3}}$ or $\left(\dfrac{3}{4}\right)^{-2}$ or $\dfrac{4}{3}$ seen **[M1]**

$\dfrac{16}{9}$ **[A1]**

(b) Writes both sides with base 2: $(2^3)^{2x-1} = (2^5)^{x+2}$ **[M1]**

Equates exponents: $3(2x-1) = 5(x+2)$ **[M1]**

$6x - 3 = 5x + 10$ **[M1]**

13 **[A1]**

[Total Marks: 6]

18 (a) Takes the reciprocal to deal with the negative index: $\left(\dfrac{64y^{12}}{27x^9}\right)^{\frac{1}{3}}$ or evaluates cube root of constants correctly **[M1]**

Applies the power of $\dfrac{1}{3}$ to variables: $y^{12\times\frac{1}{3}}$ and $x^{9\times\frac{1}{3}}$ **[M1]**

$\dfrac{4y^4}{3x^3}$ **[A1]**

(b) Writes 32 as 2^5 **[B1]**

Equates indices: $3x - 1 = 5x$ **[M1]**

-0.5 or $-\dfrac{1}{2}$ **[A1]**

[Total Marks: 6]

19 (a) 8 **[B1]**

x^9 **[B1]**

y^6 as denominator or y^{-6} **[B1]**

(b) $8 = 2^3$ seen or implied by $8^x = 2^{3x}$ **[M1]**

$x + 3 = 3x$ **[M1]**

1.5 or $\frac{3}{2}$ **[A1]**

[Total Marks: 6]

20 (a) $(3^2)^{2x} = (3^3)^{x+1}$ or $3^{4x} = 3^{3x+3}$ **[M1]**

$4x = 3x + 3$ **[M1]**

3 **[A1]**

(b) Inverts fraction to remove negative power: $\left(\frac{81b^{12}}{16a^8}\right)^{\frac{3}{4}}$ **[M1]**

Applies power $\frac{1}{4}$ (fourth root) to all terms: $\frac{3b^3}{2a^2}$ **[M1]**

Cubes the result to get final answer: $\frac{27b^9}{8a^6}$ or $3.375b^9a^{-6}$ **[A1]**

[Total Marks: 6]

21 (a) Expressing both sides with base 2: $(2^4)^{2x-1} = (2^3)^{x+2}$ **[M1]**

Equating powers: $4(2x - 1) = 3(x + 2)$ or $8x - 4 = 3x + 6$ **[M1]**

2 **[A1]**

(b) Taking the reciprocal to remove negative index: $\left(\frac{16y^8}{81x^{12}}\right)^{\frac{3}{4}}$ **[M1]**

Evaluating the fractional power of the numbers: $16^{\frac{3}{4}} = 8$ and $81^{\frac{3}{4}} = 27$ **[M1]**

$\frac{8y^6}{27x^9}$ **[A1]**

[Total Marks: 6]

22 (a) Simplifies the first fraction to $8x^2y^{-5}$ or $\frac{8x^2}{y^5}$ **[M1]**

Multiplies by the reciprocal of the second fraction: $\times\frac{y^4}{4x^3}$ **[M1]**

$\frac{2}{xy}$ or $2x^{-1}y^{-1}$ **[A1]**

(b) Writes 25 as 5^2 and 125 as 5^3 **[M1]**

Applies exponent rules correctly to get a single base on the left: $5^{2x-1+2(x+2)}$ **[M1]**

Equates indices to form a linear equation: $4x + 3 = 3x$ **[M1]**

-3 **[A1]**

[Total Marks: 7]

23 **(a)** $12x^{\frac{3}{2}}$ seen as numerator **[M1]**

$2x^k$ for any k or $cx^{\frac{7}{6}}$ for any c **[M1]**

$2x^{\frac{7}{6}}$ **[A1]**

(b) Converting all terms to base 3: $3^{2(2x-1)} \times 3^{x+1} = 3^{3x}$ **[M1]**

Applying exponent multiplication law: $3^{4x-2+x+1}$ or 3^{5x-1} **[M1]**

Equating indices: $5x - 1 = 3x$ **[M1]**

0.5 or $\frac{1}{2}$ **[A1]**

[Total Marks: 7]

24 **(a)** $(9x^{-4})^{\frac{1}{2}} = 3x^{-2}$ **[B1]**

Numerator simplified to $12x^{-\frac{1}{2}}$ **[M1]**

Subtracts indices correctly: $-\frac{1}{2} - \left(-\frac{3}{2}\right) = 1$ **[M1]**

$2x$ **[A1]**

(b) Expresses all terms as powers of 3: $\frac{3^{2k-1}}{3^{3k}} = 3^{\frac{1}{2}}$ **[M1]**

Equates indices: $(2k-1) - 3k = \frac{1}{2}$ or $-k - 1 = \frac{1}{2}$ **[M1]**

$-\frac{3}{2}$ or -1.5 **[A1]**

[Total Marks: 7]

25 **(a)** M1 for simplifying the numerator: $30x^{\frac{1}{2}}$

M1 for correctly handling the division: $6x^{\frac{1}{2}-\frac{3}{2}}$

$6x^{-1}$ or $\frac{6}{x}$ **[A1]**

(b) M1 for reciprocating to eliminate the negative index: $x^{\frac{2}{3}} = \frac{9}{4}$

M1 for isolating x: $x = \left(\frac{9}{4}\right)^{\frac{3}{2}}$

$\frac{27}{8}$ or $3\frac{3}{8}$ **[A1]**

(c) M1 for converting to base 3: $3^{2x} \times 3^{2x-1} = 3^{3(x+1)}$

M1 for applying index laws on the LHS to get 3^{4x-1}

M1 for equating powers correctly: $4x - 1 = 3x + 3$

4 **[A1]**

[Total Marks: 10]

26 **(a)** Any two of 10, x^6, y^2 correct **[M1]**

$10x^6y^2$ **[A1]**

(b) $\frac{3a^2}{b^3}$ or $3a^2b^{-3}$ **[B2]**

If 0 scored, award 1 mark for any two of 3, a^2, b^3 (or b^{-3}) correct in a product or quotient **[M1]**

(c) $\frac{1}{4}$ or 0.25 **[B1]**

[Total Marks: 5]

27 **(a)** M1 for $16^{\frac{3}{4}} = 8$ seen

M1 for $x^{12\times\frac{3}{4}} = x^9$ or $y^{8\times\frac{3}{4}} = y^6$ seen

$\frac{8x^9}{y^6}$ or $8x^9y^{-6}$ **[A1]**

(b) M1 for expressing both sides with a common base of 2: $2^{2(2x-1)}$ and $2^{3(x+2)}$

M1 for equating the exponents: $4x - 2 = 3x + 6$

8 **[A1]**

[Total Marks: 6]

28 **(a)** Expressing both sides with a base of 2: $2^{2(2x-1)}$ and 2^{5x} **[M1]**

Equating indices: $4x - 2 = 5x$ **[M1]**

$x = -2$ **[A1]**

(b) Any two of 4, x^3, y^{-4} (or $\frac{1}{y^4}$) correct **[M1]**

$4x^3y^{-4}$ or $\frac{4x^3}{y^4}$ **[A1]**

(c) $x = 64^{\frac{2}{3}}$ or $(\sqrt[3]{64})^2$ or 4^2 **[M1]**

16 **[A1]**

[Total Marks: 7]

29 **(a)** $4x^2$ or y^{-3} or $\frac{1}{y^3}$ seen **[M1]**

$\frac{4x^2}{y^3}$ or $4x^2y^{-3}$ **[A1]**

(b) 3^{2x} or $3^{3(x-1)}$ seen **[B1]**

$3\sqrt{3} = 3^{1.5}$ or $3^{\frac{3}{2}}$ seen **[B1]**

$2x + 3x - 3 = 1.5$ or equivalent equation in x **[M1]**

$x = 0.9$ or $\frac{9}{10}$ **[A1]**

[Total Marks: 6]

30 (a) Expressing all terms with a base of 2, e.g. $2^{2x} \times 2^{3(2x-1)} = 2^4$ **[M1]**

Correct linear equation for the exponents: $2x + 6x - 3 = 4$ or $8x = 7$ **[M1]**

$\frac{7}{8}$ or 0.875 **[A1]**

(b) Simplifying the numerator to $24x^1$ (or $24x$) **[M1]**

Correct manipulation of the coefficients $\frac{24}{8} = 3$ or indices $x^{1-(-3)}$ **[M1]**

$3x^4$ **[A1]**

[Total Marks: 6]

31 (a) $\left(\frac{27y^6}{8x^9}\right)^{\frac{2}{3}}$ or $\frac{4x^6}{9y^4}$ (inverted or squared/cubed root correctly applied to some terms) **[M1]**

$\left(\frac{3y^2}{2x^3}\right)^2$ **[M1]**

$\frac{9y^4}{4x^6}$ or $2.25y^4x^{-6}$ **[A1]**

(b) $\frac{5^{a-2} \times 5^{2b}}{5^{3(a+1)}} = 5^4$ (expressing all terms as powers of 5) **[M1]**

$a - 2 + 2b - 3(a + 1) = 4$ **[M1]**

$-2a + 2b - 5 = 4$ or $-2a + 2b = 9$ **[A1]**

Accept any pair of values satisfying the equation, e.g. $a = 0, b = 4.5$ or $a = -4.5, b = 0$ or $b = a + 4.5$ **[A1]**

[Total Marks: 7]

32 (a) $3x^2y^{-6}$ or $\frac{3x^2}{y^6}$ **[B3]**

Coefficient 3 seen **[M1]**

Correct subtraction of indices $x^{\frac{3}{2}-\left(-\frac{1}{2}\right)}$ leading to x^2 OR y^{-2-4} leading to y^{-6} **[M1]**

(b) Expresses all terms as powers of 2: $(2^2)^{2x-1} \times (2^3)^{x+1} = (2^5)^x$ **[M1]**

Correctly adds powers on the LHS: $2(2x-1)+3(x+1)$ or $4x-2+3x+3$ **[M1]**

Forms linear equation equating LHS and RHS powers: $7x+1=5x$ **[M1]**

$-\frac{1}{2}$ or -0.5 **[A1]**

[Total Marks: 7]

33 (a) $\frac{2}{5}a^4b^{-6}$ or $\frac{2a^4}{5b^6}$ (correctly simplifying the first fraction) **[B1]**

$\left(\frac{2}{a^2b}\right)^2$ or $\frac{4}{a^4b^2}$ or $4a^{-4}b^{-2}$ (correctly simplifying the second fraction) **[B2]**

If 0 scored for the second fraction, award 1 mark for applying the negative index: $\frac{a^{-4}b^{-2}}{2^{-2}}$ **[M1]**

$\frac{8}{5}b^{-8}$ or $\frac{8}{5b^8}$ **[A1]**

(b) $3^{2x}\times 3^y=3^7$ yielding $2x+y=7$ **[B1]**

$2^{3x}\div 2^{2y}=2^7$ yielding $3x-2y=7$ **[B1]**

Correct method to solve their simultaneous linear equations, e.g., substituting $y=7-2x$ to get $3x-2(7-2x)=7$ or matching coefficients **[M1]**

$7x=21 \implies x=3$ **[A1]**

$y=1$ **[A1]**

[Total Marks: 9]